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INTRALUMINAL LONGITUDINAL PHYSIOLOGY WEARABLE FOR TIME-SERIES TISSUE-ADJACENT MONITORING (NON-LIMITING)

A. Technical Field

This section relates to intraluminal wearable sensing devices configured to collect longitudinal time-series tissue-adjacent physiology and to computing systems configured to perform validity gating, artifact rejection, and privacy-preserving metric generation from such signals. The disclosed embodiments further relate to optional correlation of derived metrics with other nodes under consent-gated configurations.

B. Summary (Non-Limiting)

In various embodiments, an intraluminal longitudinal wearable is configured for prolonged wear to collect time-series signals over hours and to generate privacy-preserving outputs representing physiologic patterns and trends. The device may be configured for intravaginal use and, in certain embodiments, may be adapted for other intraluminal contexts. The wearable includes one or more sensing modalities including, without limitation, bio-impedance (including multi-frequency impedance), capacitive wetness/dielectric sensing, solid-state pH sensing (including ISFET pH in certain embodiments), temperature sensing, pelvic activity sensing, and distributed pressure/contact sensing. A computing system receives signals from the wearable, computes one or more quality indices, selects valid windows, and generates privacy-preserving outputs comprising indices, zones, trend curves, distributions, and confidence indicators.

C. Exemplary Physical Embodiments and Form Factors (Non-Limiting)

In various embodiments, the intraluminal wearable comprises a sealed body configured to provide stable coupling to internal tissue during movement and prolonged wear. Non-limiting form factors include: 1. an elongated soft-shell body with one or more sensing regions distributed along its length; 2. an hourglass or gently contoured body configured to promote stable placement; and/or 3. a modular body with a central electronics region and distributed sensing regions. In certain embodiments, the device includes a retrieval feature (e.g., flexible tab or loop) and a charging interface (e.g., dock-based charging). In certain embodiments, the device includes a compliant outer layer (e.g., silicone or other biocompatible elastomer) molded over an internal frame to maintain shape while remaining comfortable.

D. Sensor Suite (Non-Limiting)

In various embodiments, the intraluminal wearable includes one or more sensors including, without limitation: 1. bio-impedance sensing, including single-frequency and/or multi-frequency impedance using one or more electrode pairs; 2. dielectric/wetness sensing, including capacitive wetness sensing and dielectric trend estimation; 3. pH sensing, including solid-state pH sensing and/or ISFET pH sensing; 4. temperature sensing, including one or more thermal sensors configured to provide context and compensation; 5. pelvic activity sensing, including distributed pressure sensing, capacitive pressure sensing, piezoresistive pressure sensing, force-sensing resistors, MEMS-based sensing, and/or multi-zone pressure arrays; and 6. inertial sensing, including a 6-axis IMU configured for motion gating, orientation estimation, and artifact rejection. Any sensor modality may be substituted with an equivalent modality providing substantially similar function. Sensors may be described as sensing regions rather than fixed placements to support design variation.

E. Collection Modes and Example Use Conditions (Non-Limiting)

In various embodiments, the wearable operates in one or more modes including continuous sampling, periodic sampling with duty cycling, burst sampling during validated windows, and event-triggered sampling based on detected transitions (e.g., wetness changes, impedance changes, activity changes, thermal changes). In certain embodiments, the device is configured for multi-hour wear sessions and stores time-stamped data on-node for later synchronization.

F. Standalone Use and Optional Integration Hooks (Non-Limiting)

In standalone embodiments, Node 2 generates privacy-preserving longitudinal outputs for a single user without requiring any other node. In optional multi-node embodiments, derived metrics from Node 2 may be time-aligned and correlated with other nodes (including external longitudinal nodes, baseline geometry/compliance nodes, and/or upper-body context nodes) under consent gating and quality thresholds.